

OpenAI Model Masters

A structured reference handbook for OpenAI models, modalities, capabilities, APIs, tools, and retrieval workflows

Catalog snapshot: 26 September 2026

Purpose: a deliberately information-dense, cross-referenced PDF designed for Retrieval-Augmented Generation (RAG) indexing practice. It is not an official OpenAI publication and should not be treated as a contractual specification.

The handbook repeats important model names, capability labels, modality terms, API concepts, and use cases across sections. This makes it useful for testing chunking, embedding, semantic retrieval, metadata filtering, reranking, citation, and question answering.

Primary source family: official OpenAI developer documentation and OpenAI model catalog.

How to Use This Corpus for RAG

This document is intentionally organized into short, independently meaningful sections. A production RAG pipeline can chunk it by heading, paragraph, page, or semantic unit.

Suggested experiments

- Index the PDF as one document and compare fixed-size chunks against heading-aware chunks.
- Ask exact model questions such as: Which model is described as optimized for cost-sensitive workloads?
- Ask cross-section questions such as: Which APIs can accept PDFs, and what happens to PDF page images?
- Test synonym retrieval: 'speech to text' versus 'transcription', 'vector index' versus 'vector store', and 'vision' versus 'image understanding'.
- Test metadata filters by model family, modality, lifecycle status, and API surface.
- Measure whether retrieval returns the correct model profile when several model names appear in the same chapter.
- Evaluate citation grounding by requiring answers to identify the relevant section title and page.

Important freshness rule

OpenAI model catalogs are dynamic. Model IDs, aliases, pricing, context limits, supported features, and deprecation status can change. The dated snapshot in this handbook is therefore part of the dataset and should be preserved in benchmark metadata.

Glossary of Core Terms

Term	Meaning in this handbook
Model	A named OpenAI model or model family exposed through an API or product surface.
Modality	A kind of information a model can receive or produce, such as text, image, or audio.
Context window	The amount of model input and related context that can be processed in a request.
Reasoning	Model computation used to work through difficult tasks before producing an answer.
Token	A unit used by language models to represent text or other serialized content.
Embedding	A numeric vector representing the semantic properties of text or another supported input.
Vector store	A searchable container for vectorized content used by retrieval systems.
Retrieval	Finding relevant indexed content for a natural-language query.
File Search	A hosted Responses API tool that searches uploaded files using semantic and keyword retrieval.
Structured Outputs	A capability for constraining model responses to a specified schema.
Function calling	A mechanism for a model to request execution of application-defined functions or tools.
Vision	The ability to analyze image inputs.
Transcription	Converting spoken audio into text.
Text-to-speech	Generating spoken audio from text.
Realtime	Low-latency interaction patterns designed for streaming or conversational applications.
Deprecated	A model or API entry that OpenAI identifies as deprecated; migration may be required.

OpenAI Model Taxonomy

The current official model catalog groups models by practical function rather than presenting one single linear hierarchy.

Category	Examples / role	Typical task
Flagship general-purpose	GPT-6 Astra, GPT-6 Sol, GPT-6 Luna	Reasoning, coding, professional work, high-volume workloads
Earlier general-purpose	GPT-5.6 family, GPT-5.x family, GPT-4.1	General generation, coding, reasoning, production workloads
Image	GPT-Image-2.5 Sunburst, GPT-Image-2.5 Flare, GPT-Image-2.5	Image generation and editing
Realtime / audio	GPT-Live 1, GPT-Realtime-2.1, GPT-Audio-1.5	Voice, realtime interaction, audio input/output
Transcription	GPT-Transcribe, GPT-4o Transcribe	Speech-to-text
Speech generation	GPT-4o Mini TTS, TTS-1, TTS-1 HD	Text-to-speech
Embeddings	text-embedding-3-small, text-embedding-3-large	Semantic search and vector representations
Moderation	omni-moderation	Text/image safety classification
Open-weight	gpt-oss-120b, gpt-oss-20b	Self-hosted or open-weight workflows where applicable
Specialized	Cyber, life-sciences and other purpose-built models	Domain-specific workflows

The catalog also contains older and deprecated entries. Lifecycle status is a first-class retrieval attribute.

The Responses API as a General Interface

The Responses API is described in OpenAI's current developer documentation as a primary interface for direct model requests, structured output, tools, and multimodal workflows.

Conceptual request

A request typically identifies a model and supplies input. Depending on the model and tool configuration, the request may include text, images, files, tool definitions, or other supported content.

Example shape:

```
client.responses.create({ model: 'gpt-6-astra', input: 'Explain retrieval augmented generation.' })
```

Important API ideas

- Model selection is explicit in production-oriented examples.
- Input can represent instructions, questions, multimodal content, or tool-related context.
- The model can return natural-language output, structured output, tool calls, and other response items depending on the workflow.
- The Responses API can expose hosted tools such as File Search and image generation.
- The API is a transport and orchestration surface; the model's actual capabilities still depend on the selected model.

RAG relevance

For a RAG application, the Responses API can sit after retrieval: retrieve relevant chunks, provide them as context, then ask a model to synthesize an answer while preserving source references.

Input and Output Modalities

A useful RAG catalog should distinguish what a model can understand from what it can produce.
'Multimodal' does not mean every model accepts every modality.

Modality	Input example	Output example	Common model category
Text	Question, instructions, code	Answer, code, JSON	General-purpose GPT
Image	PNG, JPEG, WEBP, GIF	Analysis or generated image	Vision and image models
PDF/file	Uploaded document	Text/structured response	Vision-capable general models via file inputs
Audio	Speech stream or audio file	Transcription or audio response	Realtime, transcription, audio models
Speech	Text prompt for voice generation	Synthesized speech	TTS models
Vector	Text to embed	Embedding vector	Embedding models

The official model catalog should be treated as the authoritative place to verify a specific model's supported input and output modalities.

Prompting and Model Behavior

OpenAI's API documentation describes prompts as instructions or examples used to program model behavior. Prompt quality is separate from model selection, but the two interact.

A robust prompt can specify

- The task to perform.
- Relevant context and source material.
- Constraints and exclusions.
- Output format or schema.
- Examples of successful behavior when examples are useful.
- How uncertainty should be expressed.
- Whether citations or source identifiers must be preserved.

For RAG

A retrieval prompt should distinguish retrieved evidence from the user's question. A useful instruction is to answer from supplied evidence, identify missing information, and avoid inventing facts that are not supported by the retrieved context.

RAG prompt pattern:

Context: [retrieved chunks]

Question: [user query]

Task: answer using the context; cite the supporting chunk identifiers; state when the context is insufficient.

Structured Outputs

Structured Outputs are intended for applications where the model response should conform to a defined structure rather than being unrestricted prose.

Use case	Potential schema fields
Model catalog extraction	model_id, family, status, modalities, reasoning, context_window
RAG retrieval result	chunk_id, document_id, relevance, evidence
Support answer	answer, confidence_note, citations
Classification	label, rationale, evidence
Agent action	action_type, arguments, approval_required

RAG benefit

Structured output makes downstream evaluation easier. A benchmark can compare expected fields rather than attempting to parse arbitrary prose.

Caution

Schema compliance does not automatically make the content factually correct. Retrieval grounding, validation, and application-level checks remain important.

Tools and Function Calling

Tool-enabled workflows let models interact with application capabilities instead of returning only prose. A tool call represents a request by the model to perform an external operation.

- Function calling can connect a model to application-defined functions.
- Hosted tools can provide capabilities managed by OpenAI.
- The application remains responsible for executing actions and enforcing permissions.
- Tool arguments should be validated before execution.
- For consequential actions, applications should use confirmation or approval controls where appropriate.

RAG connection

Retrieval itself can be exposed as a tool. The model first decides that it needs evidence, calls a retrieval function, receives relevant chunks, and then composes an answer.

Tool loop: user question → model → retrieval tool → ranked chunks → model → grounded answer

File Inputs

The Responses API supports file inputs through `input_file` items. OpenAI's documentation distinguishes PDF handling from other document types.

File category	General processing described by OpenAI
PDF	For vision-capable models, text extraction and page images can both be provided.
DOC/DOCX/RTF/ODT	Text extraction for input processing.
PPT/PPTX	Document text extraction for supported file input workflows.
TXT/MD/JSON/HTML/XML/code	Text-oriented processing.
CSV/XLS/XLSX/TSV	Spreadsheet-specific augmentation is available for supported workflows.

A current documentation note says a single file included in a request must be under 50 MB, with a 50 MB combined request limit for file inputs in the described workflow. Limits can change and should be rechecked before production.

PDF Inputs and Visual Detail

PDFs are especially useful in multimodal retrieval experiments because page images can preserve visual information that plain text extraction might lose.

The Responses API documentation describes a PDF detail setting with auto, low, and high options. The setting controls page-image processing detail; extracted text remains part of PDF processing.

Detail	Conceptual purpose
auto	Use the model's default behavior for page-image detail.
low	Reduce visual input cost when fine visual detail is not needed.
high	Preserve more visual information for dense pages, small print, or diagrams.

RAG experiment

Index this handbook as text and separately test direct PDF understanding. Compare retrieval from extracted text against a vision model's ability to inspect page layout.

Images and Vision

Vision-capable models can analyze image inputs. OpenAI documentation describes use cases such as describing images, reading visible text, and answering questions about objects, shapes, colors, or textures.

Supported image examples

- PNG.
- JPEG or JPG.
- WEBP.
- Non-animated GIF.
- Multiple images can be supplied in one request when supported by the endpoint and model.

Vision is distinct from image generation. A vision-capable language model can analyze an image; an image-generation model produces or edits images.

Vision pipeline: image → model analysis → text/structured response

Generation pipeline: text + optional image context → image model → image output

Image Generation Models

The current catalog lists GPT-Image-2.5 Sunburst as a highly capable image generation and editing model, GPT-Image-2.5 Flare as a faster everyday image model, and GPT-Image-2 as a state-of-the-art image generation model.

Model	Catalog role	Primary modality
GPT-Image-2.5 Sunburst	Most capable image generation/editing in the listed family	Text/image → image
GPT-Image-2.5 Flare	Fast, high-quality everyday image generation	Text/image → image
GPT-Image-2	State-of-the-art image generation model	Text/image → image

When documenting a model, record generation capability separately from image understanding. They are related but not interchangeable capabilities.

Audio: Realtime, Transcription, and Speech

OpenAI's model catalog separates realtime/audio interaction, transcription, and speech generation into distinct categories.

Workflow	Input	Output	Representative models
Realtime voice	Streaming audio/text	Streaming audio/text	GPT-Live 1, GPT-Realtime-2.1
Transcription	Audio	Text	GPT-Transcribe, GPT-4o Transcribe
Speech generation	Text	Audio	GPT-4o Mini TTS, TTS-1, TTS-1 HD
Speech-to-speech	Audio	Audio	GPT-Audio-1.5 / realtime audio models

The exact endpoint, streaming behavior, latency, and modality support depend on the specific model and API surface.

Transcription Models

Transcription models convert spoken audio into text. The catalog currently lists GPT-Transcribe, GPT-Live-Transcribe, GPT-Realtime-Whisper, GPT-4o Transcribe, GPT-4o Mini Transcribe, and a diarization-oriented GPT-4o Transcribe variant.

Model	Description in catalog
GPT-Transcribe	High-accuracy speech-to-text for file and realtime input transcription.
GPT-Live-Transcribe	Low-latency speech-to-text for realtime transcription.
GPT-Realtime-Whisper	Streaming speech-to-text for realtime transcription.
GPT-4o Transcribe	Speech-to-text model powered by GPT-4o.
GPT-4o Mini Transcribe	Smaller speech-to-text model powered by GPT-4o Mini.
GPT-4o Transcribe Diarize	Transcription model that identifies who is speaking when.

RAG use case

Meeting audio can be transcribed first, segmented into speaker-aware chunks, embedded, and indexed for semantic search.

Text-to-Speech Models

Text-to-speech systems convert written text into spoken audio. The catalog includes both current and legacy entries for speech generation.

Model	Catalog description / role
GPT-4o Mini TTS	Text-to-speech model powered by GPT-4o Mini.
TTS-1	Text-to-speech model optimized for speed.
TTS-1 HD	Text-to-speech model optimized for quality.

For application design, separate three decisions: the semantic model that writes text, the speech model that voices it, and the transport used to stream or deliver audio.

Embeddings and Semantic Representation

OpenAI embeddings convert text into numeric vectors. Similar texts tend to have vectors that are closer according to a chosen similarity measure.

Model	Documented role	Default vector length
text-embedding-3-small	Smaller, efficient embedding model	1536
text-embedding-3-large	Most capable current embedding model	3072
text-embedding-ada-002	Older embedding model	1536

The newer embedding models support a `dimensions` parameter, allowing applications to shorten vectors to trade some representation capacity for lower storage and compute requirements.

RAG pipeline

chunk text → embedding model → vector → vector database/store → similarity search → retrieved chunks → generation model

Retrieval and Vector Stores

OpenAI's Retrieval API describes semantic search as finding content that is semantically similar to a natural-language query, even when few keywords overlap.

Vector stores act as indexes for searchable data. When files are added, the service can chunk, embed, and index them for retrieval.

Stage	Meaning
Ingest	Add documents/files to the retrieval system.
Chunk	Split source content into retrievable units.
Embed	Represent chunks as vectors.
Index	Store vectors and searchable metadata.
Query	Convert the user question into a retrieval operation.
Rank	Return the most relevant chunks.
Generate	Use retrieved evidence to answer the question.

File Search

File Search is a hosted tool in the Responses API. It lets a model search uploaded files for relevant information before generating a response.

OpenAI describes File Search as combining semantic and keyword search over files associated with vector stores.

Feature	RAG significance
Vector store	Persistent searchable container for indexed files.
Semantic search	Retrieves conceptually related content.
Keyword search	Helps with exact terms and names.
Result limit	Can constrain how many results are retrieved.
Search result inclusion	Can be requested when the application needs returned retrieval evidence.

This handbook itself is suitable for File Search experiments because the same concepts occur across model-specific and API-specific sections.

Moderation Models

OpenAI's moderation API is designed to identify potentially harmful content in text and images. The current catalog identifies omni-moderation as a moderation model.

Capability	Description
Text classification	Classifies potentially harmful text.
Image classification	Can classify supported image inputs.
Audio	The cited moderation documentation says omni-moderation does not classify audio.
Application policy	Results can be used for filtering, routing, review, or intervention.

Moderation is not a general-purpose answer model. It is a specialized classification component.

Computer Use

Computer-use workflows allow a model to operate browser or desktop interfaces through application-provided execution mechanisms.

The current documentation describes two broad execution approaches: code execution using tools such as PyAutoGUI or Playwright, and a computer tool that returns structured mouse and keyboard actions for an application to execute.

- The model observes screenshots and tool results.
- The application controls the actual environment.
- UI actions should be validated and permissioned.
- Computer use is different from ordinary text generation because the model participates in an action loop.

RAG connection

An agent can retrieve documentation, inspect a UI, and then use retrieved evidence to decide what UI action to request. Retrieval and computer use therefore solve different parts of the workflow.

Agents and Model Selection

Agentic applications combine models with tools, state, instructions, and execution loops. OpenAI documentation recommends explicit model selection in production workflows rather than relying on an implicit runtime default.

Decision factor	Question
Quality	How much reasoning and capability does this task need?
Latency	How quickly must the application respond?
Cost	What input/output price and workload volume are acceptable?
Tool support	Does the selected model support the required tool features?
Modalities	Does it accept and/or produce the required media?
Lifecycle	Is the model current, deprecated, or scheduled for migration?

Model Profile: GPT-6 Astra

Role: Flagship general-purpose reasoning model

The current catalog describes GPT-6 Astra as OpenAI's most capable model, built for the hardest end-to-end work. Guidance emphasizes computer use, browsing, software engineering, science, and professional workflows.

Field	Reference value
Model ID / name	GPT-6 Astra
Primary inputs	Text; multimodal workflows where supported
Representative capabilities	Reasoning, coding, agents, professional work, computer use
Lifecycle/catalog status	Current flagship
Retrieval keywords	GPT-6 Astra; Flagship general-purpose reasoning model; Reasoning, coding, agents, professional work, computer use

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-6 Sol

Role: General-purpose complex coding and agentic model

The catalog describes GPT-6 Sol as built to power complex coding and agentic workflows. It occupies the high-capability tier below Astra in the current catalog presentation.

Field	Reference value
Model ID / name	GPT-6 Sol
Primary inputs	Text and supported multimodal inputs
Representative capabilities	Coding, agents, complex workflows
Lifecycle/catalog status	Current flagship family
Retrieval keywords	GPT-6 Sol; General-purpose complex coding and agentic model; Coding, agents, complex workflows; OpenAI n

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-6 Luna

Role: Efficient general-purpose model

GPT-6 Luna is described as the most efficient GPT-6 option for focused, high-volume tasks. It is intended for workloads where throughput, latency, and cost sensitivity matter.

Field	Reference value
Model ID / name	GPT-6 Luna
Primary inputs	Text and supported inputs
Representative capabilities	Focused tasks, high volume, cost-sensitive applications
Lifecycle/catalog status	Current flagship family
Retrieval keywords	GPT-6 Luna; Efficient general-purpose model; Focused tasks, high volume, cost-sensitive applications; OpenAI

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.6 Sol

Role: Flagship professional-work model

The catalog describes GPT-5.6 Sol as a flagship model for complex professional work. It belongs to the GPT-5.6 generation and supports production-oriented reasoning and generation workflows.

Field	Reference value
Model ID / name	GPT-5.6 Sol
Primary inputs	Text and supported multimodal inputs
Representative capabilities	Professional work, complex tasks
Lifecycle/catalog status	Current family
Retrieval keywords	GPT-5.6 Sol; Flagship professional-work model; Professional work, complex tasks; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.6 Terra

Role: Balanced GPT-5.6 model

GPT-5.6 Terra is described as balancing intelligence and cost. It is useful as a catalog example of a model positioned between maximum capability and cost-sensitive deployment.

Field	Reference value
Model ID / name	GPT-5.6 Terra
Primary inputs	Text and supported inputs
Representative capabilities	Balanced production workloads
Lifecycle/catalog status	Current family
Retrieval keywords	GPT-5.6 Terra; Balanced GPT-5.6 model; Balanced production workloads; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.6 Luna

Role: Cost-sensitive GPT-5.6 model

GPT-5.6 Luna is described as optimized for cost-sensitive workloads. It is useful for high-volume tasks where the application needs GPT-5.6-family capabilities without the highest-tier model economics.

Field	Reference value
Model ID / name	GPT-5.6 Luna
Primary inputs	Text and supported inputs
Representative capabilities	High-volume, cost-sensitive workloads
Lifecycle/catalog status	Current family
Retrieval keywords	GPT-5.6 Luna; Cost-sensitive GPT-5.6 model; High-volume, cost-sensitive workloads; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.5

Role: Coding and professional-work model

The catalog describes GPT-5.5 as a new class of intelligence for coding and professional work. It is part of the broader GPT-5.x model lineage.

Field	Reference value
Model ID / name	GPT-5.5
Primary inputs	Text and supported inputs
Representative capabilities	Coding, professional work
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5.5; Coding and professional-work model; Coding, professional work; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.5 Pro

Role: Higher-precision GPT-5.5 variant

GPT-5.5 Pro is described as a version of GPT-5.5 that produces smarter and more precise responses. It represents a higher-capability variant within the family.

Field	Reference value
Model ID / name	GPT-5.5 Pro
Primary inputs	Text and supported inputs
Representative capabilities	Professional work, higher precision
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5.5 Pro; Higher-precision GPT-5.5 variant; Professional work, higher precision; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.4

Role: Affordable coding/professional model

GPT-5.4 is described as a more affordable model for coding and professional work. It is positioned for applications needing strong capability with more cost awareness.

Field	Reference value
Model ID / name	GPT-5.4
Primary inputs	Text and supported inputs
Representative capabilities	Coding, professional work
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5.4; Affordable coding/professional model; Coding, professional work; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.4 Pro

Role: Higher-precision GPT-5.4 variant

GPT-5.4 Pro is described as a version of GPT-5.4 that produces smarter and more precise responses.

Field	Reference value
Model ID / name	GPT-5.4 Pro
Primary inputs	Text and supported inputs
Representative capabilities	Professional work, higher precision
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5.4 Pro; Higher-precision GPT-5.4 variant; Professional work, higher precision; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.4 Mini

Role: Compact coding/computer-use model

The catalog describes GPT-5.4 Mini as a strong mini model for coding, computer use, and subagents.

Field	Reference value
Model ID / name	GPT-5.4 Mini
Primary inputs	Text and supported inputs
Representative capabilities	Coding, computer use, subagents
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5.4 Mini; Compact coding/computer-use model; Coding, computer use, subagents; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.4 nano

Role: Low-cost high-volume model

GPT-5.4 nano is described as the cheapest GPT-5.4-class model for simple, high-volume tasks.

Field	Reference value
Model ID / name	GPT-5.4 nano
Primary inputs	Text and supported inputs
Representative capabilities	Simple high-volume workloads
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5.4 nano; Low-cost high-volume model; Simple high-volume workloads; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.3-Codex

Role: Agentic coding model

The catalog describes GPT-5.3-Codex as the most capable agentic coding model to date in its listing. It is optimized for long-horizon coding workflows.

Field	Reference value
Model ID / name	GPT-5.3-Codex
Primary inputs	Text/code and supported inputs
Representative capabilities	Agentic coding, long-horizon development
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5.3-Codex; Agentic coding model; Agentic coding, long-horizon development; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.2

Role: Professional reasoning model

GPT-5.2 is described as a previous flagship model for professional work with configurable reasoning effort.

Field	Reference value
Model ID / name	GPT-5.2
Primary inputs	Text and supported inputs
Representative capabilities	Professional work, reasoning
Lifecycle/catalog status	Previous/current catalog entry
Retrieval keywords	GPT-5.2; Professional reasoning model; Professional work, reasoning; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.2 Pro

Role: Higher-precision GPT-5.2 variant

GPT-5.2 Pro is described as a previous professional-work model that produces smarter and more precise responses.

Field	Reference value
Model ID / name	GPT-5.2 Pro
Primary inputs	Text and supported inputs
Representative capabilities	Professional work
Lifecycle/catalog status	Previous/current catalog entry
Retrieval keywords	GPT-5.2 Pro; Higher-precision GPT-5.2 variant; Professional work; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5.1

Role: Coding and agentic model

GPT-5.1 is described as a strong model for coding and agentic tasks with configurable reasoning effort.

Field	Reference value
Model ID / name	GPT-5.1
Primary inputs	Text and supported inputs
Representative capabilities	Coding, agents
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5.1; Coding and agentic model; Coding, agents; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5

Role: Reasoning model for coding and agents

GPT-5 is described as an intelligent reasoning model for coding and agentic tasks with configurable reasoning effort.

Field	Reference value
Model ID / name	GPT-5
Primary inputs	Text and supported inputs
Representative capabilities	Coding, agents, reasoning
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5; Reasoning model for coding and agents; Coding, agents, reasoning; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5 Mini

Role: Cost-sensitive reasoning model

GPT-5 Mini is described as providing strong intelligence for cost-sensitive, low-latency, high-volume workloads.

Field	Reference value
Model ID / name	GPT-5 Mini
Primary inputs	Text and supported inputs
Representative capabilities	High-volume, low-latency workloads
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5 Mini; Cost-sensitive reasoning model; High-volume, low-latency workloads; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5 nano

Role: Fastest low-cost GPT-5 variant

GPT-5 nano is described as the fastest, most cost-efficient version of GPT-5.

Field	Reference value
Model ID / name	GPT-5 nano
Primary inputs	Text and supported inputs
Representative capabilities	Simple, high-volume tasks
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5 nano; Fastest low-cost GPT-5 variant; Simple, high-volume tasks; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-5 Pro

Role: Higher-precision GPT-5 variant

GPT-5 Pro is described as a version of GPT-5 that produces smarter and more precise responses.

Field	Reference value
Model ID / name	GPT-5 Pro
Primary inputs	Text and supported inputs
Representative capabilities	Complex professional tasks
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-5 Pro; Higher-precision GPT-5 variant; Complex professional tasks; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: o3

Role: Earlier reasoning model

o3 is listed as a reasoning model for complex tasks and is identified in the current catalog as having been succeeded by GPT-5.

Field	Reference value
Model ID / name	o3
Primary inputs	Text and supported inputs
Representative capabilities	Complex reasoning
Lifecycle/catalog status	Previous generation
Retrieval keywords	o3; Earlier reasoning model; Complex reasoning; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: o3-pro

Role: Higher-compute o3 variant

o3-pro is described as a version of o3 with more compute for better responses.

Field	Reference value
Model ID / name	o3-pro
Primary inputs	Text and supported inputs
Representative capabilities	Complex reasoning
Lifecycle/catalog status	Previous generation
Retrieval keywords	o3-pro; Higher-compute o3 variant; Complex reasoning; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-4.1

Role: Non-reasoning general model

GPT-4.1 is described as the smartest non-reasoning model in the current catalog.

Field	Reference value
Model ID / name	GPT-4.1
Primary inputs	Text, vision where supported
Representative capabilities	General generation, coding, vision
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-4.1; Non-reasoning general model; General generation, coding, vision; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-4.1 Mini

Role: Smaller GPT-4.1 variant

GPT-4.1 Mini is described as a smaller, faster version of GPT-4.1.

Field	Reference value
Model ID / name	GPT-4.1 Mini
Primary inputs	Text, vision where supported
Representative capabilities	General tasks, lower latency
Lifecycle/catalog status	Current/available catalog entry
Retrieval keywords	GPT-4.1 Mini; Smaller GPT-4.1 variant; General tasks, lower latency; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-4o

Role: General-purpose multimodal model

GPT-4o is listed as a fast, intelligent, flexible GPT model. The family is also associated with image and audio-capable variants.

Field	Reference value
Model ID / name	GPT-4o
Primary inputs	Text and image; audio variants exist
Representative capabilities	General tasks, vision, multimodal applications
Lifecycle/catalog status	Current/legacy catalog entry
Retrieval keywords	GPT-4o; General-purpose multimodal model; General tasks, vision, multimodal applications; OpenAI model; AI

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-4o Mini

Role: Small general-purpose model

GPT-4o Mini is described as a fast, affordable small model for focused tasks.

Field	Reference value
Model ID / name	GPT-4o Mini
Primary inputs	Text and supported multimodal inputs
Representative capabilities	Focused, cost-sensitive tasks
Lifecycle/catalog status	Current/legacy catalog entry
Retrieval keywords	GPT-4o Mini; Small general-purpose model; Focused, cost-sensitive tasks; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Realtime-2.1

Role: Realtime reasoning model

The catalog describes GPT-Realtime-2.1 as a reasoning model with tool use for realtime applications.

Field	Reference value
Model ID / name	GPT-Realtime-2.1
Primary inputs	Realtime audio/text as supported
Representative capabilities	Voice agents, realtime tool use
Lifecycle/catalog status	Current realtime family
Retrieval keywords	GPT-Realtime-2.1; Realtime reasoning model; Voice agents, realtime tool use; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Realtime-2.1 Mini

Role: Smaller realtime reasoning model

GPT-Realtime-2.1 Mini is listed as a smaller realtime reasoning model with tool use.

Field	Reference value
Model ID / name	GPT-Realtime-2.1 Mini
Primary inputs	Realtime audio/text as supported
Representative capabilities	Realtime agents, lower-cost workloads
Lifecycle/catalog status	Current realtime family
Retrieval keywords	GPT-Realtime-2.1 Mini; Smaller realtime reasoning model; Realtime agents, lower-cost workloads; OpenAI mo

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Realtime-Translate

Role: Realtime speech translation

GPT-Realtime-Translate is described as a streaming speech-to-speech translation model.

Field	Reference value
Model ID / name	GPT-Realtime-Translate
Primary inputs	Streaming speech
Representative capabilities	Realtime translation
Lifecycle/catalog status	Current realtime family
Retrieval keywords	GPT-Realtime-Translate; Realtime speech translation; Realtime translation; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Realtime-1.5

Role: Realtime audio model

GPT-Realtime-1.5 is described as the best voice model for audio in and audio out in the current catalog listing.

Field	Reference value
Model ID / name	GPT-Realtime-1.5
Primary inputs	Audio in/out
Representative capabilities	Voice conversations
Lifecycle/catalog status	Current/available realtime family
Retrieval keywords	GPT-Realtime-1.5; Realtime audio model; Voice conversations; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Audio-1.5

Role: Audio input/output model

GPT-Audio-1.5 is described as the best model for audio in and audio out with Chat Completions.

Field	Reference value
Model ID / name	GPT-Audio-1.5
Primary inputs	Audio in/out
Representative capabilities	Audio applications
Lifecycle/catalog status	Current audio family
Retrieval keywords	GPT-Audio-1.5; Audio input/output model; Audio applications; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Transcribe

Role: High-accuracy transcription

GPT-Transcribe is described as a high-accuracy speech-to-text model for file and realtime input transcription.

Field	Reference value
Model ID / name	GPT-Transcribe
Primary inputs	Audio input
Representative capabilities	Transcription
Lifecycle/catalog status	Current transcription family
Retrieval keywords	GPT-Transcribe; High-accuracy transcription; Transcription; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-4o Transcribe

Role: GPT-4o speech-to-text

GPT-4o Transcribe is described as a speech-to-text model powered by GPT-4o.

Field	Reference value
Model ID / name	GPT-4o Transcribe
Primary inputs	Audio input
Representative capabilities	Transcription
Lifecycle/catalog status	Current/available transcription family
Retrieval keywords	GPT-4o Transcribe; GPT-4o speech-to-text; Transcription; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-4o Mini Transcribe

Role: Smaller transcription model

GPT-4o Mini Transcribe is described as a speech-to-text model powered by GPT-4o Mini.

Field	Reference value
Model ID / name	GPT-4o Mini Transcribe
Primary inputs	Audio input
Representative capabilities	Cost-sensitive transcription
Lifecycle/catalog status	Current/available transcription family
Retrieval keywords	GPT-4o Mini Transcribe; Smaller transcription model; Cost-sensitive transcription; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-4o Transcribe Diarize

Role: Speaker-aware transcription

This transcription model is described as identifying who is speaking when, adding speaker diarization to transcription workflows.

Field	Reference value
Model ID / name	GPT-4o Transcribe Diarize
Primary inputs	Audio input
Representative capabilities	Transcription with speaker identification
Lifecycle/catalog status	Current/available transcription family
Retrieval keywords	GPT-4o Transcribe Diarize; Speaker-aware transcription; Transcription with speaker identification; OpenAI mo

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Image-2.5 Sunburst

Role: Advanced image generation/editing

The catalog describes GPT-Image-2.5 Sunburst as the most capable model for image generation and editing.

Field	Reference value
Model ID / name	GPT-Image-2.5 Sunburst
Primary inputs	Text/image input
Representative capabilities	Image generation and editing
Lifecycle/catalog status	Current image family
Retrieval keywords	GPT-Image-2.5 Sunburst; Advanced image generation/editing; Image generation and editing; OpenAI model; A

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Image-2.5 Flare

Role: Fast image generation

GPT-Image-2.5 Flare is described as a fast, high-quality everyday image generation model.

Field	Reference value
Model ID / name	GPT-Image-2.5 Flare
Primary inputs	Text/image input
Representative capabilities	Everyday image generation
Lifecycle/catalog status	Current image family
Retrieval keywords	GPT-Image-2.5 Flare; Fast image generation; Everyday image generation; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: GPT-Image-2

Role: State-of-the-art image generation

GPT-Image-2 is listed as a state-of-the-art image generation model.

Field	Reference value
Model ID / name	GPT-Image-2
Primary inputs	Text/image input
Representative capabilities	Image generation
Lifecycle/catalog status	Current image family
Retrieval keywords	GPT-Image-2; State-of-the-art image generation; Image generation; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: text-embedding-3-large

Role: Advanced embedding model

text-embedding-3-large is described as the most capable embedding model. It creates vector representations for semantic search and related applications.

Field	Reference value
Model ID / name	text-embedding-3-large
Primary inputs	Text input
Representative capabilities	Embeddings, retrieval, clustering
Lifecycle/catalog status	Current embedding family
Retrieval keywords	text-embedding-3-large; Advanced embedding model; Embeddings, retrieval, clustering; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: text-embedding-3-small

Role: Efficient embedding model

text-embedding-3-small is described as a smaller, efficient embedding model with strong multilingual performance and adjustable dimensions.

Field	Reference value
Model ID / name	text-embedding-3-small
Primary inputs	Text input
Representative capabilities	Embeddings, retrieval, classification
Lifecycle/catalog status	Current embedding family
Retrieval keywords	text-embedding-3-small; Efficient embedding model; Embeddings, retrieval, classification; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: omni-moderation

Role: Safety classification

omni-moderation is used to identify potentially harmful content in text and images. It is a specialized moderation component rather than a general answer model.

Field	Reference value
Model ID / name	omni-moderation
Primary inputs	Text and image
Representative capabilities	Moderation
Lifecycle/catalog status	Current moderation family
Retrieval keywords	omni-moderation; Safety classification; Moderation; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: gpt-oss-120b

Role: Open-weight model

gpt-oss-120b is listed as the most powerful OpenAI open-weight model and is described as fitting into an H100 GPU.

Field	Reference value
Model ID / name	gpt-oss-120b
Primary inputs	Text and supported inputs
Representative capabilities	Open-weight deployment
Lifecycle/catalog status	Open-weight
Retrieval keywords	gpt-oss-120b; Open-weight model; Open-weight deployment; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Profile: gpt-oss-20b

Role: Medium open-weight model

gpt-oss-20b is listed as a medium-sized open-weight model intended for low latency.

Field	Reference value
Model ID / name	gpt-oss-20b
Primary inputs	Text and supported inputs
Representative capabilities	Open-weight, low latency
Lifecycle/catalog status	Open-weight
Retrieval keywords	gpt-oss-20b; Medium open-weight model; Open-weight, low latency; OpenAI model; API

Verification note

This profile is a dated catalog snapshot. For production integration, verify the model's current documentation page, supported endpoints, limits, pricing, and lifecycle status before implementation.

Model Selection by Workload

Model Selection by Workload

Key	Content
Complex end-to-end work	GPT-6 Astra — High capability and broad professional/agentic focus
Complex coding and agents	GPT-6 Sol — Coding and agentic workflow focus
Focused high-volume work	GPT-6 Luna — Efficiency and cost-sensitive positioning
Image generation/editing	GPT-Image-2.5 family — Purpose-built image generation/editing
Realtime voice	GPT-Live / GPT-Realtime family — Streaming audio and realtime interaction
Transcription	GPT-Transcribe family — Speech-to-text
Embeddings	text-embedding-3-small / large — Vector representations for search and retrieval
Moderation	omni-moderation — Text/image safety classification

RAG note

These entries are intentionally explicit and repetitive so that a retrieval system can be tested for exact-match, semantic-match, metadata filtering, and multi-hop synthesis behavior.

Model Lifecycle and Deprecation

Key	Content
Current	Available model listed without a deprecated label — Verify current documentation
Previous generation	Still listed but described as succeeded by newer models — Plan migrations where appropriate
Deprecated	Catalog explicitly labels the entry deprecated — Do not build new dependencies without a migration plan
Alias	A name may point to a model family or changing implementation — Pin explicit model IDs when stability matters

RAG note

These entries are intentionally explicit and repetitive so that a retrieval system can be tested for exact-match, semantic-match, metadata filtering, and multi-hop synthesis behavior.

RAG Evaluation Questions

Key	Content
Exact retrieval	Which model is described as optimized for cost-sensitive workloads?
Modality	Which OpenAI documentation distinguishes PDF text extraction from PDF page images?
Embeddings	What are the default dimensions listed for text-embedding-3-small and text-embedding-3-large?
File Search	What does File Search use vector stores for?
Moderation	What input modalities does omni-moderation accept according to the cited guide?
Computer use	What are the two broad execution approaches described for computer use?
Audio	Which model category is used for speech-to-text?
Image	How is image understanding different from image generation?
Lifecycle	Which model entries are explicitly described as deprecated?
API	Why might a production system select an explicit model instead of a runtime default?

RAG note

These entries are intentionally explicit and repetitive so that a retrieval system can be tested for exact-match, semantic-match, metadata filtering, and multi-hop synthesis behavior.

RAG Chunk Metadata Schema

Key	Content
document_id	Stable identifier for this PDF or source document
page	PDF page number
section	Heading under which the chunk appears
model	Specific model name if the chunk describes one
family	Model family such as GPT-6, GPT-5.x, realtime, image, embedding
modality	Input/output modality labels
lifecycle	Current, previous, deprecated, open-weight, specialized
source_type	Official OpenAI documentation, generated corpus section, benchmark
snapshot_date	2026-09-26
keywords	Names, aliases, capability terms, endpoint terms

RAG note

These entries are intentionally explicit and repetitive so that a retrieval system can be tested for exact-match, semantic-match, metadata filtering, and multi-hop synthesis behavior.

Benchmark Query Set: Model Identity

Key	Content
Q1	What is GPT-6 Astra?
Q2	What is GPT-6 Luna optimized for?
Q3	Which model is described as built for complex coding and agentic workflows?
Q4	What is GPT-5.4 Mini intended for?
Q5	What is GPT-4.1's position in the catalog?
Q6	Which model is described as a previous flagship for professional work?
Q7	What is GPT-5 nano designed to optimize?
Q8	What is GPT-5.3-Codex specialized for?
Q9	Which model is described as a speaker-aware transcription model?
Q10	Which model family is intended for image generation?

RAG note

These entries are intentionally explicit and repetitive so that a retrieval system can be tested for exact-match, semantic-match, metadata filtering, and multi-hop synthesis behavior.

Benchmark Query Set: Capabilities

Key	Content
Q11	Which OpenAI models can analyze image inputs?
Q12	What file formats can the Responses API accept as input files?
Q13	How are PDFs processed differently from ordinary text documents?
Q14	What is semantic retrieval?
Q15	What is a vector store?
Q16	What does File Search do?
Q17	What are embeddings used for?
Q18	What is Structured Outputs?
Q19	What does computer use allow a model to do?
Q20	What does moderation classify?

RAG note

These entries are intentionally explicit and repetitive so that a retrieval system can be tested for exact-match, semantic-match, metadata filtering, and multi-hop synthesis behavior.

Benchmark Query Set: Cross-Document Reasoning

Key	Content
Q21	How would you build a meeting-search RAG system using transcription and embeddings?
Q22	When would a developer use File Search instead of directly passing a large file as input?
Q23	How are image understanding and image generation different API tasks?
Q24	Why should model lifecycle status be stored as retrieval metadata?
Q25	How can structured outputs improve RAG evaluation?
Q26	Why should retrieved evidence be separated from user instructions in a RAG prompt?
Q27	What information should be recorded for a model catalog entry?
Q28	How can a RAG system retrieve by model name while also filtering by lifecycle status?
Q29	How can embedding dimensions affect retrieval infrastructure cost?
Q30	Why does a model catalog need a snapshot date?

RAG note

These entries are intentionally explicit and repetitive so that a retrieval system can be tested for exact-match, semantic-match, metadata filtering, and multi-hop synthesis behavior.

Primary Sources

The following official OpenAI documentation pages were consulted when constructing this dated reference corpus. URLs are included inside the artifact so the source trail remains part of the indexed content.

Primary Sources

The following official OpenAI documentation pages were consulted when constructing this dated reference corpus. URLs are included inside the artifact so the source trail remains part of the indexed content.

OpenAI API — All models

Current model catalog and lifecycle labels.

<https://developers.openai.com/api/docs/models/all>

OpenAI API — Compare models

Model capability and specification comparison.

<https://developers.openai.com/api/docs/models/compare>

OpenAI API Documentation

API surface, build paths, models and workflows.

<https://developers.openai.com/api/docs>

OpenAI API — Key concepts

Prompting, text generation, privacy and core concepts.

<https://developers.openai.com/api/docs/concepts>

OpenAI API — File inputs

File types, PDF processing and input limits.

<https://developers.openai.com/api/docs/guides/file-inputs>

OpenAI API — Images and vision

Image understanding and image generation workflows.

<https://developers.openai.com/api/docs/guides/images-vision>

OpenAI API — Retrieval

Semantic search, vector stores, chunking and embeddings.

<https://developers.openai.com/api/docs/guides/retrieval>

OpenAI API — File Search

Hosted file retrieval through the Responses API.

<https://developers.openai.com/api/docs/guides/tools-file-search>

OpenAI API — Vector embeddings

Embedding models and semantic representation.

<https://developers.openai.com/api/docs/guides/embeddings>

OpenAI API — Moderation

Text/image moderation.

<https://developers.openai.com/api/docs/guides/moderation>

OpenAI API — Structured Outputs

Schema-constrained model output.

<https://developers.openai.com/api/docs/guides/structured-outputs>

OpenAI API — Computer use

Computer interaction workflows.

<https://developers.openai.com/api/docs/guides/tools-computer-use>

OpenAI API — Model guidance

Current-family model guidance and migration context.

<https://developers.openai.com/api/docs/guides/latest-model>

Corpus Design Notes

This PDF was generated specifically for RAG indexing practice. It intentionally mixes narrative explanations, tables, model profiles, terminology, benchmark questions, metadata fields, and source references.

Recommended ingestion metadata

- document_id = openai-model-masters-2026-09-26
- document_type = synthetic_reference_handbook
- topic = openai_models_and_api
- snapshot_date = 2026-09-26
- source_authority = official_openai_documentation
- language = en

Important distinction

This artifact is a practice corpus, not a replacement for live OpenAI documentation. If an implementation decision depends on current limits, pricing, model availability, or endpoint support, verify against the current official documentation.

RAG Appendix 1: Retrieval Practice Vocabulary

This appendix intentionally repeats related terminology with varied phrasing to create additional semantic neighborhoods for embedding experiments.

Concept	Related expressions
Model	LLM, language model, GPT model, model ID, model family
Retrieval	semantic search, similarity search, knowledge retrieval, evidence lookup
Chunk	segment, passage, retrievable unit, text span
Embedding	vector representation, semantic vector, dense representation
Vector store	vector index, semantic index, retrieval store
Multimodal	text and image, audio and text, visual input, multimodal model
Transcription	speech-to-text, ASR, audio transcription
Generation	text generation, image generation, speech synthesis
Tool use	function calling, hosted tools, agent actions
Grounding	evidence-based answer, source grounding, retrieval grounding

Practice query

Find all passages that discuss semantic search, vector stores, embeddings, and File Search. Then compare whether the retriever favors exact terminology or conceptual similarity.

RAG Appendix 2: Retrieval Practice Vocabulary

This appendix intentionally repeats related terminology with varied phrasing to create additional semantic neighborhoods for embedding experiments.

Concept	Related expressions
Model	LLM, language model, GPT model, model ID, model family
Retrieval	semantic search, similarity search, knowledge retrieval, evidence lookup
Chunk	segment, passage, retrievable unit, text span
Embedding	vector representation, semantic vector, dense representation
Vector store	vector index, semantic index, retrieval store
Multimodal	text and image, audio and text, visual input, multimodal model
Transcription	speech-to-text, ASR, audio transcription
Generation	text generation, image generation, speech synthesis
Tool use	function calling, hosted tools, agent actions
Grounding	evidence-based answer, source grounding, retrieval grounding

Practice query

Find all passages that discuss semantic search, vector stores, embeddings, and File Search. Then compare whether the retriever favors exact terminology or conceptual similarity.

RAG Appendix 3: Retrieval Practice Vocabulary

This appendix intentionally repeats related terminology with varied phrasing to create additional semantic neighborhoods for embedding experiments.

Concept	Related expressions
Model	LLM, language model, GPT model, model ID, model family
Retrieval	semantic search, similarity search, knowledge retrieval, evidence lookup
Chunk	segment, passage, retrievable unit, text span
Embedding	vector representation, semantic vector, dense representation
Vector store	vector index, semantic index, retrieval store
Multimodal	text and image, audio and text, visual input, multimodal model
Transcription	speech-to-text, ASR, audio transcription
Generation	text generation, image generation, speech synthesis
Tool use	function calling, hosted tools, agent actions
Grounding	evidence-based answer, source grounding, retrieval grounding

Practice query

Find all passages that discuss semantic search, vector stores, embeddings, and File Search. Then compare whether the retriever favors exact terminology or conceptual similarity.

RAG Appendix 4: Retrieval Practice Vocabulary

This appendix intentionally repeats related terminology with varied phrasing to create additional semantic neighborhoods for embedding experiments.

Concept	Related expressions
Model	LLM, language model, GPT model, model ID, model family
Retrieval	semantic search, similarity search, knowledge retrieval, evidence lookup
Chunk	segment, passage, retrievable unit, text span
Embedding	vector representation, semantic vector, dense representation
Vector store	vector index, semantic index, retrieval store
Multimodal	text and image, audio and text, visual input, multimodal model
Transcription	speech-to-text, ASR, audio transcription
Generation	text generation, image generation, speech synthesis
Tool use	function calling, hosted tools, agent actions
Grounding	evidence-based answer, source grounding, retrieval grounding

Practice query

Find all passages that discuss semantic search, vector stores, embeddings, and File Search. Then compare whether the retriever favors exact terminology or conceptual similarity.